

24-502-0107: Quantum Computing

Core/Elective: **Elective**

Semester: **1**

Credits: **3**

Course Description

This course introduces concepts of Quantum Computing. This includes representation of quantum information and use of quantum algorithms. The course will cover design of quantum circuits and provide understanding of quantum noise and quantum cryptography.

Course Outcomes (CO)

After the completion of the course, the students will be able to:

CO	Course Outcome Statement	Cognitive Level
CO1	Understand the basic concepts of Quantum Computing.	Understand
CO2	Examine quantum representation of information and encryption.	Analyze
CO3	Design quantum circuits using qubit gates.	Apply
CO4	Apply quantum algorithms for computation.	Apply
CO5	Understand quantum noise and error correction for fault tolerant computation.	Understand

Mapping with Program Outcomes

	PO1	PO2	PO3	PO4	PSO1	PSO2	PSO3	PSO4
CO1	2	1	-	-	2	-	-	-
CO2	1	-	-	-	1	1	1	-
CO3	2	2	-	2	2	1	-	1
CO4	1	2	-	1	1	-	-	-
CO5	2	1	-	2	2	2	1	1

Course Content

1. Introduction to Quantum Computation: Quantum bits, representation of a qubit and multiple qubits. Quantum mechanics, Probabilities and measurements, entanglement, density operators and correlation, Measurements in bases other than computational basis.
2. Quantum correlations: Bell inequalities and entanglement, Schmidt decomposition, super-dense coding, teleportation, PPT criterion. Quantum Circuits: single qubit operations, multiple qubit gates, Universal quantum gates, design of quantum circuits.
3. Quantum Algorithms: Classical computation on quantum computers. Relationship between quantum and classical complexity classes. Deutsch's algorithm, Deutsch's-Jozsa algorithm, quantum search.
4. Quantum Information and Cryptography: Comparison between classical and quantum information theory. Shannon entropy, noiseless coding, Bell states. Quantum teleportation. Quantum Cryptography, no cloning theorem.
5. Noise and error correction: Quantum noise and quantum operations, Flip code, The Shor code, Quantum error correction, Stabilizer codes, Hamming bound. fault-tolerant quantum computation.

References

1. Quantum Computation and Quantum Information, M. A. Nielsen & I. Chuang, Cambridge University Press, 10th Edition, 2010.
2. J. Hidary, Quantum Computing: An Applied Approach, 1st Edition, Springer Publishing, 2019.
3. David McMahon, Quantum computing explained, Wiley-Interscience, 1st Edition, John Wiley & Sons, Inc. Publication, 2008.
4. A. Peres. Quantum Theory: Concepts and Methods. 1st Edition, Springer, 1995.
5. J. Preskill, Lecture Notes on Quantum Information and Computation, California Institute of Technology, 2021.
6. Mark M. Wilde, Quantum information Theory, 1st Edition, Cambridge University Press, 2012.
7. D. A. Lidar and T. A. Brun, Quantum error correction, 1st Edition, Cambridge University Press, 2013.